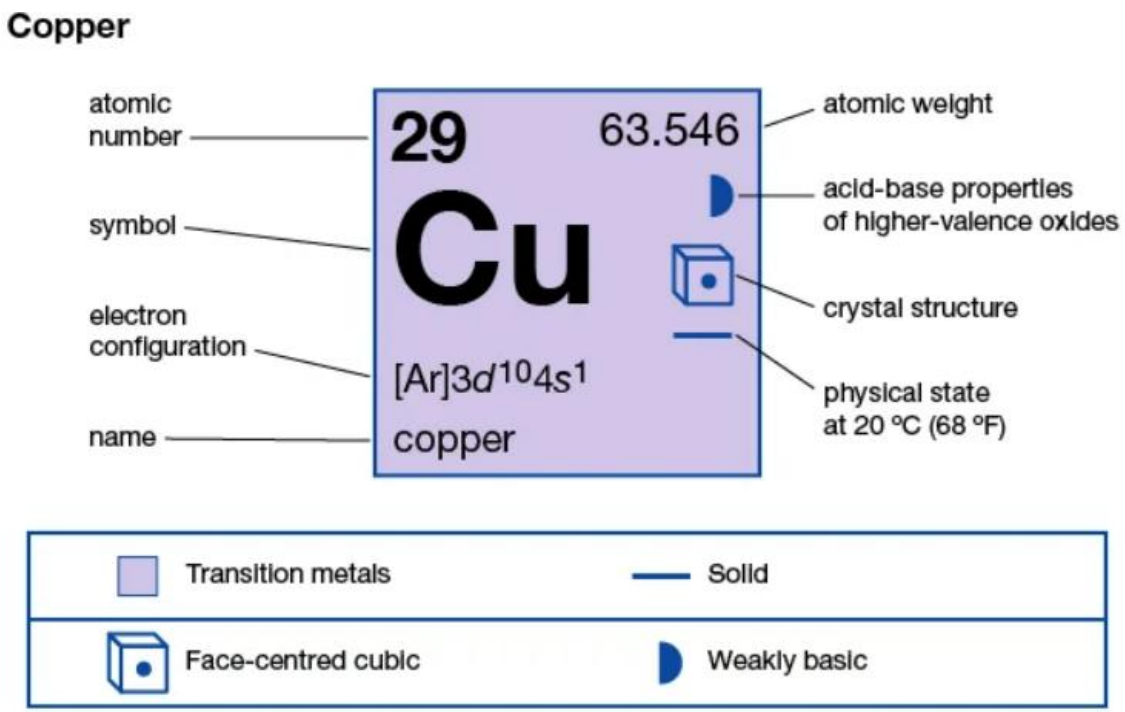
Heavy Metals

- The term heavy metals refers to any metals that has a relatively high density i.e. greater than 5 g/cc. E.g. : Copper, Iron, Gold, Silver, Mercury, Lead, Zinc etc.

Copper (तामा)
(Cu-Latin: cuprum as extracted from island of cuprus)

Introduction



- Belongs to group 11 (IB) with Silver (Ag) and Gold (Au).
- They are called **coinage metals** as they had been used for making coins as currency in past.
- They all lie below hydrogen in electrochemical series, hence can't liberate hydrogen gas from dilute acids.
- Reactivity order as reducing agent: Cu > Ag > Au

Ores: Ores are naturally occurring rocks or minerals from which metals can be extracted conveniently and profitably.

Ores of Copper

1.	Copper pyrite (chalcopyrite)	$CuFeS_2$ (chief ore)
2.	Cuprite	Cu_2O
3.	Chalcocite (copper glance)	Cu_2S
4.	Malachite	$Cu(OH)_2 \cdot CuCO_3$
5.	Azurite	$[Cu(OH)_2 \cdot 2CuCO_3]$

Copper pyrite

Cuprite

Chalcocite

Malachite

Azurite

Extraction of Copper from Copper Pyrites

The main ore of the copper is copper pyrites. Mostly, cooper is extracted from copper pyrite using dry process (pyrometallurgy).

Pyrometallurgy: the process of extracting metals from their ores using high temperature (oxides, sulphides)

Theory/Principle:

Copper is extracted from copper pyrite by oxidation and self-reduction. On roasting, cuprous sulphide (Cu_2S) and ferrous sulphide (FeS) are obtained. The FeS is oxidized to FeO, which reacts with silica flux to form slag ($FeSiO_3$), while part of Cu_2S is oxidized to Cu_2O. The Cu_2O then reacts with the remaining Cu_2S in a self-reduction reaction to give metallic copper. The crude blister copper obtained is finally purified by electrolytic refining.

Oxidation: Roasting

$$2CuFeS_2 + O_2 \rightarrow Cu_2S + 2FeS + SO_2$$

$$2FeS + 3O_2 \rightarrow 2FeO + 2SO_2$$

$$2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2$$

Slag formation:

$$FeO + SiO_2 \rightarrow FeSiO_3 \text{ (slag)}$$

Self-reduction:

$$Cu_2O + Cu_2S \rightarrow 6Cu + SO_2$$

Extraction of Copper

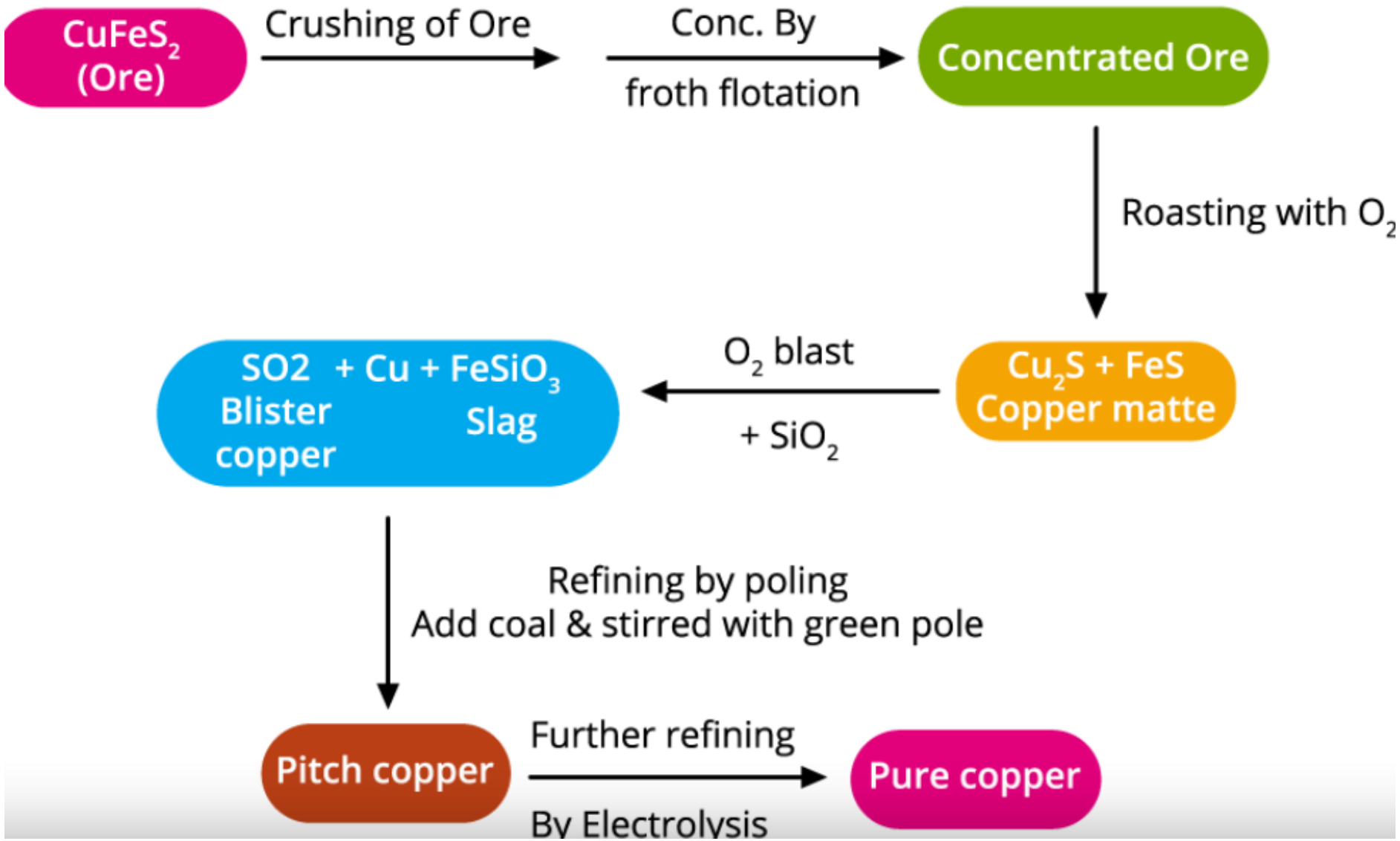


Figure: Flowsheet diagram for the extraction of copper from copper pyrite ore

Steps Involved in Metallurgy

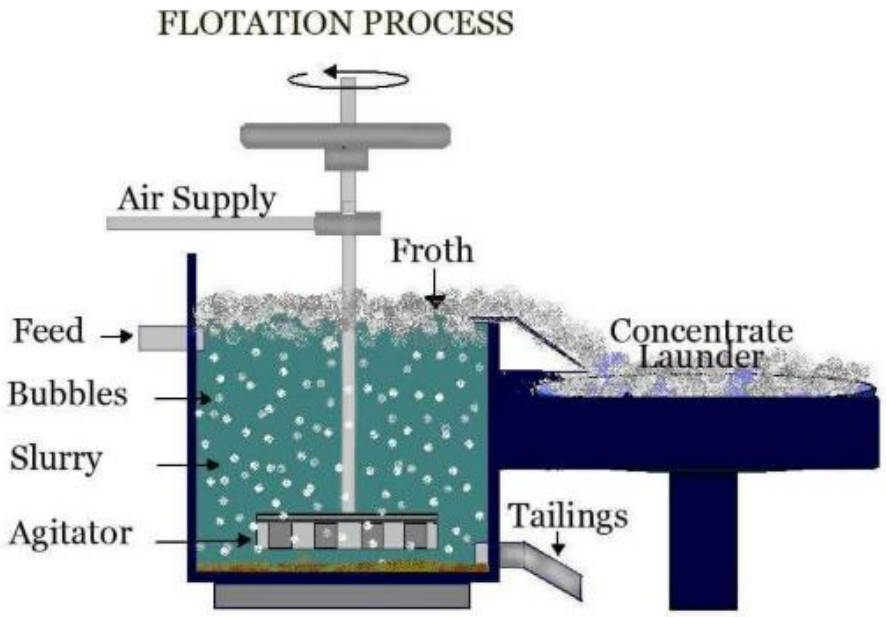
Metallurgy: Science of extracting metals from ores and preparing them for use.

1. Crushing and pulverization:

- Big lumps of ores are crushed using jaw crushers to get crushed ore which is pulverized (powdered) using a pulverizer or stamp mill to get powdered ore.
- Purpose: To increase the surface area for the following processes.

2. Concentration of ores:

- Process: Froth flotation. Separation is based on difference in wetting properties.
- Ore is mixed with water and pine oil, and air is blown.
- Sulphide ore sticks to froth (wetted by oil), while impurities (gangue) sink down (wetted by water).
- The froth is skimmed off to collect the concentrated ore.



froth flotation machines

Imp

3. Roasting:

- Concentrated ore is heated in the presence of oxygen in reverberatory furnace.
- Changes that takes place:

❖Removal of moistures and volatile substances.

❖Decomposition of copper pyrite into cuprous sulphide and ferrous sulphide.

$$2CuFeS_2 \rightarrow Cu_2S + 2FeS + SO_2$$

❖Oxidation of Cuprous and ferrous sulphides

$$2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2$$

$$2FeS + 3O_2 \rightarrow 2FeO + 2SO_2$$

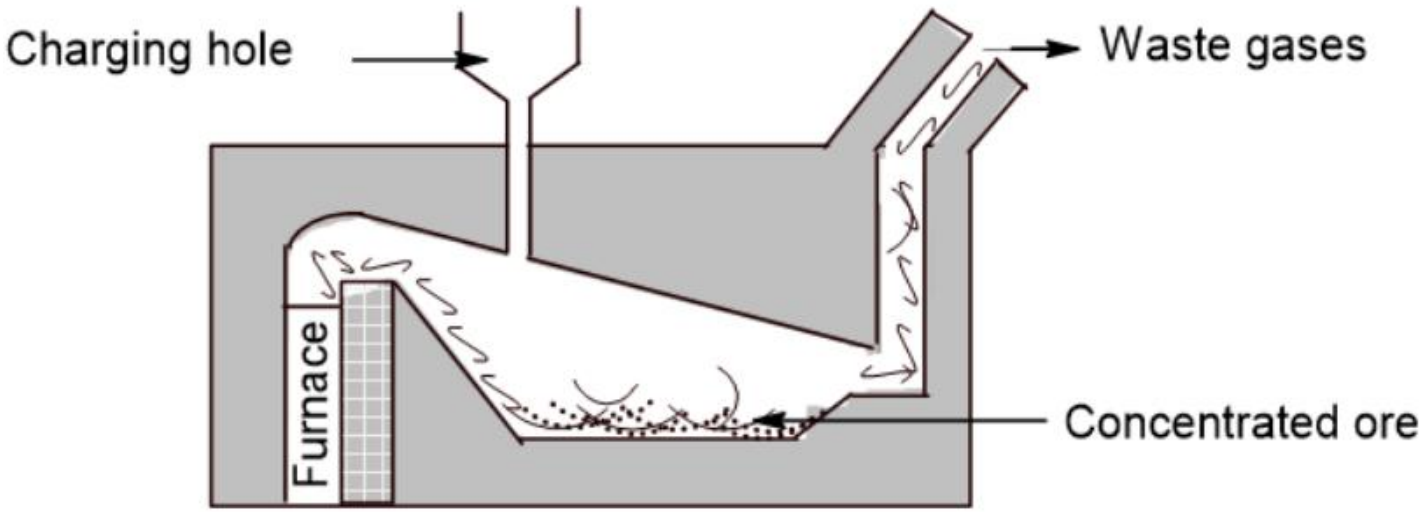


Fig: Reverberatory furnace

Imp

4. Smelting: (heating an ore in the presence of a suitable flux and a reducing agent to obtain the metal in molten form)

- The roasted ore is heated in a **blast furnace** (15-20 feet height and 6 feet in diameter) mixing with coke (fuel) and silica (SiO_2 - flux).
- A blast of air is passed from the lower part of the furnace.
- Following changes takes place:

❖ Oxidation of unreacted FeS into FeO:- $2FeS + 3O_2 \rightarrow 2FeO + 2SO_2$

❖ Combination of remaining ~~Cu_2O~~ FeS with ~~FeS~~ Cu_2O to form Cu_2S:- $Cu_2O + FeS \rightarrow Cu_2S + FeO$

❖ Conversion of FeO into slag by the action of sand (silica). Slag being light and molten, removed from the upper layer:- $FeO + SiO_2(flux) \rightarrow FeSiO_3(slag)$

- Hearth of the furnace contain molten mass containing **50% of copper and mixture of Cu_2S and FeS** called **copper matte**

Why blast furnace is called so?

Ans: It is because a **strong blast of hot air** is blown into it to maintain high temperature for smelting.

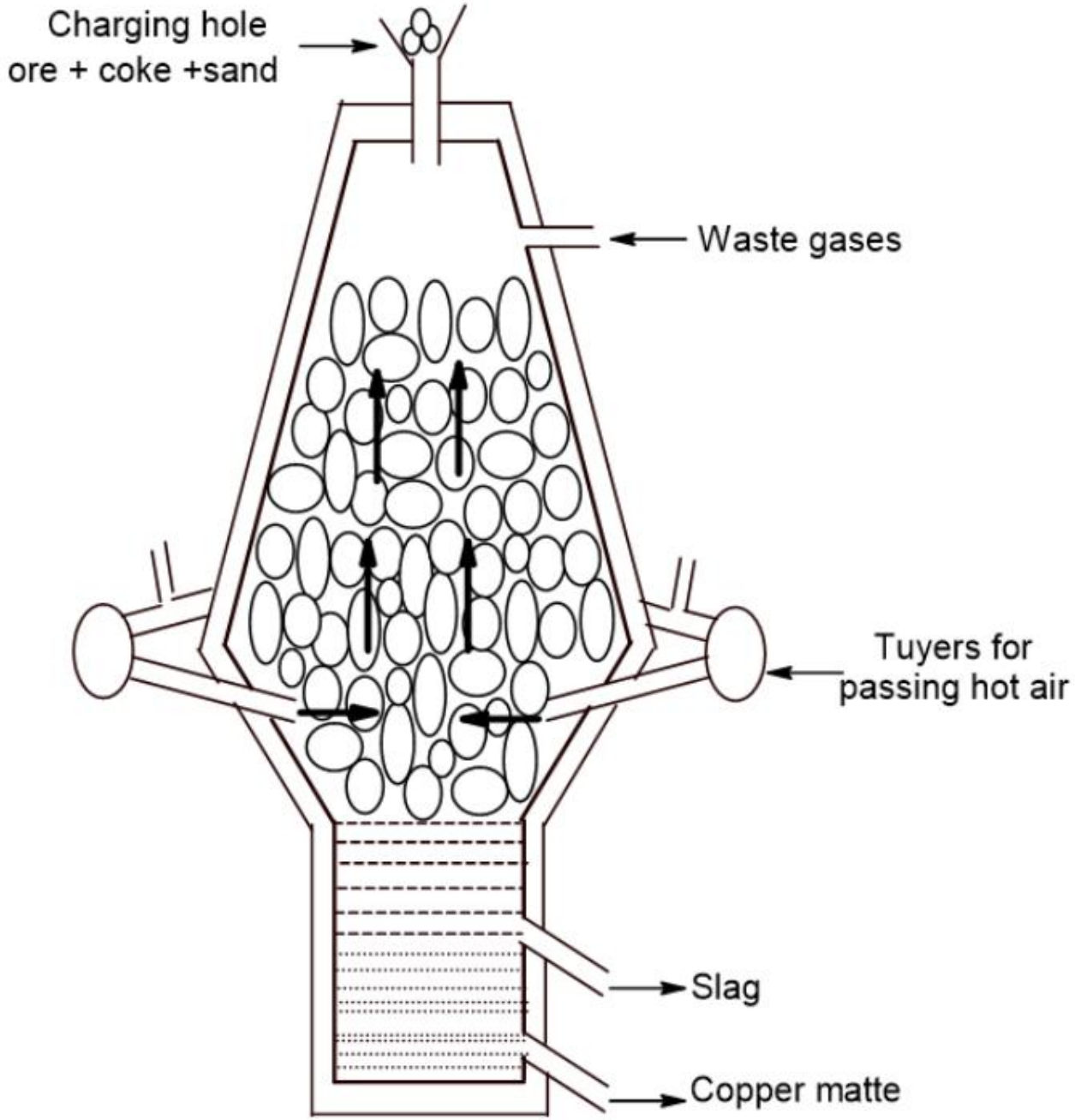


Fig: Blast furnace

5. Bessemerization: (converting matte to blister copper) – (Imp)

- Carried out in Bessemer convertor where molten copper matte obtained from smelting process is transferred.
- The blast of hot air is passed through tuyers while air is blown through molten matte
- Following changes occurs:

❖ Oxidation of FeS into FeO, which is removed as slag combining with silica. The slag is removed from top of molten mass while SO2 escapes out as waste gas.

$$2FeS + 3O_2 \rightarrow 2FeO + 2SO_2$$

$$FeO + SiO_2(flux) \rightarrow FeSiO_3(slag)$$

❖ Oxidation of Cu2S into Cu2O, which reacts with remaining Cu2S to give free copper metal **(self-reduction)**

$$2Cu_2S + 3O_2 \rightarrow 2Cu_2O + 2SO_2$$

$$Cu_2S + 2Cu_2O \rightarrow 6Cu + SO_2$$

- The product is **blister copper** (≈98% pure, with blisters due to escaping SO_2)

6. Purification or Refining:

- Obtained blister copper may still contain impurities. They can be purified by two methods:

i. **Poling (as wooden poles are used):** Blister copper may contain Cu2O which makes the copper impure and brittle. Hence, the blister copper is stirred with pole of green wood containing hydrocarbon with reduces Cu2O into metallic copper. Obtained Cu is 99.5% pure.

$$Cu_2O + C(\text{from green wood}) \rightarrow 2Cu + CO$$

ii. **Electro refining (process of purifying metals by using electrolysis, where the impure metal is taken as the anode, a thin sheet of pure metal as the cathode, and a suitable electrolyte solution of the metal salt. On passing electricity, pure metal gets deposited on the cathode while impurities are left behind or settle as anode mud):** Anode (+) is impure copper obtained from above process, cathode (-) is thin pure copper sheet and the electrolytic solution is acidified copper sulphate solution. On passing DC current through the cell: $CuSO_4 \rightleftharpoons Cu^{2+} + SO_4^{2-}$

At anode:- copper is dissolved as: $Cu \rightarrow Cu^{2+} + 2e^-$

At cathode:- Copper ions are deposited as pure copper as: $Cu^{2+} + 2e^- \rightarrow Cu$

Impurities are left below the anode as **anode mud.** Obtained copper is 99.95% pure.

Copper continued…

Chemical Properties of Copper

1. Action with air:

- No reaction with dry air but on exposure to moist air containing carbondioxide, copper gets slowly converted into green basic copper carbonate.

$$2Cu + CO_2 + H_2O + O_2 \rightarrow \underset{\substack{Basic\ copper \\ carbonate(Green)}}{CuCO_3.Cu(OH)_2}$$

- When heated with air, it forms black oxide (cupric oxide) below 1100 °C whereas red oxide (cuprous oxide) above 1100 °C.

$$2Cu + O_2 \xrightarrow{below\ 1100^{\circ}C} \underset{\substack{Cupric\ oxide \\ (black)}}{2\overset{+2}{Cu}O} \qquad 4Cu + O_2 \xrightarrow{above\ 1100^{\circ}C} \underset{\substack{Cuprous\ oxide \\ (red)}}{2\overset{+1}{Cu_2}O}$$

2. Action with acids:

- Since, copper lies below hydrogen in electrochemical series, it cannot displace hydrogen gas from dilute and non-oxidizing acids.
- Copper can react with dilute and non-oxidizing acids (dil. HCl and H_2SO_4) only in presence of air.

$$2Cu + 2HCl + O_2 \rightarrow 2CuCl_2 + H_2O \qquad 2Cu + 2H_2SO_4 + O_2 \rightarrow 2CuSO_4 + 2H_2O$$

- With conc. H_2SO_4, it gives copper sulphate and SO_2.

$$Cu + 2H_2SO_4(conc.) \rightarrow CuSO_4 + SO_2 + 2H_2O$$

- With nitric acid, copper can dissolve in different concentration of nitric acid to give copper nitrate and nitrogen oxide where the nature of nitrogen oxide depends upon the strength of HNO_3.

$$Cu + 4HNO_3(conc.) \rightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$$

$$3Cu + 8HNO_3(dil.) \rightarrow 3Cu(NO_3)_2 + 2NO + 4H_2O$$

3. Action with ammonia solution:

- Copper can react with aqueous ammonia in presence of air and forms the complex compound (Tetraamminecopper(II) hydroxide).

$$2Cu + 8NH_3 + 2H_2O + O_2 \rightarrow [Cu(NH_3)_4](OH)_2$$

4. Displacement reaction (with metal ion):

- Copper can displace less electropositive metals (which lies below it in electrochemical series) like Ag from their water-soluble salt.

$$Cu + 2AgNO_3 \rightarrow Cu(NO_3)_2 + 2Ag$$

5. Reducing property:

- Copper can reduce ferric salt to ferric salt behaving as a reducing agent.

ferrous

$$Cu + Fe_2(SO4)_3 \longrightarrow 2FeSO_4 + CuSO_4$$

$$Cu + 2FeCl_3 \longrightarrow 2FeCl_2 + CuCl_2$$

Uses of Copper:

- It is used for making utensils, electrical wires, cables, coins etc.
- It is used in electroplating and electrotyping.
- It is used to prepare various types of alloys.

Alloys of Copper:

Alloys	Composition	Uses
Brass	Cu (70%) + Zn (≈30%)	Used for making screws, nuts, bolts, musical instruments, decorative items, and utensils.
Bronze	Cu(≈80%) + Sn (≈20%)	Used for making coins, statues, medals, gears, and bearings.
Gun metal	Cu(87%) + Sn (≈10%) + Zn (≈3%)	Used for making guns, machine parts
German Silver (Nickel Silver)	Cu(50%) + Zn (30%) + Ni (20%)	Used for making ornaments, cutlery, musical instruments, and electrical fittings.

Compounds of Copper

Blue Vitriol (Nilo Tootho)

- Formula: $CuSO_4.5H_2O$
- Pentahydrate copper sulphate
- Soluble in water
- Blue in color

Preparation of blue vitriol:

1. From Cu, CuO, $Cu(OH)_2$, and $CuCO_3$ with dil H_2SO_4 :

When copper (in presence of oxygen), CuO, $Cu(OH)_2$ or $CuCO_3$ is treated with dil. H_2SO_4 and the resulting solution is crystallized, the blue vitriol is obtained.

$$2Cu + O_2 + \text{dil. } 2H_2SO_4 \longrightarrow 2CuSO_4 + 2H_2O$$

$$CuO + \text{dil. } H_2SO_4 \longrightarrow CuSO_4 + H_2O$$

$$CuCO_3 + H_2SO_4 \rightarrow CuSO_4 + H_2O + CO_2$$

$$Cu(OH)_2 + H_2SO_4 \rightarrow CuSO_4 + 2H_2O$$

$$CuSO_4(aq) \xrightarrow{crystallization} CuSO_4.5H_2O$$

Preparation of blue vitriol:

Imp

2. From scrap copper with dil H_2SO_4 :

On treating **scrap copper (waste/discarded copper materials)** with dil. H_2SO_4 in presence of air followed by the crystallization, blue vitriol is obtained. This method is used in the industry for the large scale production of blue vitriol.

$$2Cu + O_2 + \text{dil. } 2H_2SO_4 \longrightarrow 2CuSO_4 + 2H_2O$$

$$CuSO_4(aq) \xrightarrow{crystallization} CuSO_4.5H_2O$$

Copper continued…

Chemical Properties of Blue Vitriol

Imp

1. Action of heat:

- The blue vitriol looses the water of crystallization on heating gently forming anhydrous $CuSO_4$ and thermal decomposition occurs on strong heating.

$$CuSO_4.5H_2O \xrightarrow{air} CuSO_4.3H_2O \xrightarrow{100^oC} CuSO_4.H_2O \xrightarrow{230\ °C} CuSO_4 \xrightarrow{750\ °C} CuO + SO_2 + O_2$$

Bluish white

white

>1100°C

$$\cancel{750\ °C}$$

$$Cu_2O + O_2$$

Red oxide

Imp

2. Action with ammonia solution:

- Dropwise (limited) amount of ammonia solution forms bluish white ppt. of Copper hydroxide while further adding (excess) forms the deep blue coloration of Tetraamminecopper(II) sulphate complex called **Schweitzer's reagent** (use to dissolve cellulose).

$$NH_3 + H_2O \rightarrow NH_4OH$$

$$CuSO_4 + 2NH_4OH \rightarrow \underset{\text{blue ppt.}}{Cu(OH)_2} + (NH_4)_2SO_4$$

$$Cu(OH)_2 + (NH_4)_2SO_4 + 2(NH_4OH) \rightarrow \underset{\text{Deep blue colour}}{[Cu(NH_3)_4]SO_4} + 4H_2O$$

Schweitzer's reagent

3. Acidic nature:

- Aqueous copper sulphate solution is acidic due to cationic hydrolysis.

$$CuSO_{4(aq)} + 2H_2O_{(l)} \rightleftharpoons Cu(OH)_2 + H_2SO_4$$

4. Oxidizing nature:

- Can oxidize KI solution into iodine while itself get reduced to cuprous iodide.
- Amount of iodine liberated is proportional to the amount of copper in copper sulphate, hence used in iodometric titration to estimate the copper.

$$CuSO_4 + 2\,KI \longrightarrow \underset{\text{Unstable}}{CuI_2} + K_2SO_4] \times 2$$

$$2\,CuI_2 \longrightarrow Cu_2I_2 + I_2$$

$$2\,CuSO_4 + 4\,KI \longrightarrow 2\,K_2SO_4 + \underset{\text{(white)}}{Cu_2I_2\downarrow} + \underset{\text{(violet)}}{I_2}$$

Uses of Blue vitriol: (Imp – at least 2)

- Used in electroplating and electrotyping.
- Used as fungicide and germicide.
- Used in preservation of wood, gum, etc.
- Used as mordant (binding/fixing agent) in dying.

Oxides of Copper

Cuprous Oxide (Cu_2O = Red Oxide)

- O.S of Cu = +1
- Red colored powder insoluble in water.

Uses:

- Used in preparation of ruby red glass.
- Used as enamel and pigment.
- Used in preparation of antirust paints.
- Used as pesticides.

Cupric Oxide (CuO = Black Oxide)

- O.S of Cu = +2
- Black colored powder insoluble in water.

Uses:

- Used for coloring glass in black or blue color.
- Used to estimate C and H in organic compounds (quantitative analysis).
- Used to remove sulphur from petrol.
- Used as an oxidizing agent.

The End